Topic: Genetic Algorithm for Optimization Problems:

Tittle: Diet and Menu Planning.

```
import random

# Food items: [cost, calories, protein]
foods = {
    "Rice":   [20, 200, 4],
    "Milk":   [30, 150, 8],
    "Egg":    [10, 70, 6],
    "Banana": [5, 100, 1]
}

TARGET_CALORIES = 1000
TARGET_PROTEIN = 40

POP_SIZE = 20
GENERATIONS = 50

# Create a chromosome
def create_chromosome():
    return [random.randint(0, 10) for _ in range(len(foods))]

# Fitness function
def fitness(chrom):
    total_cost = 0
    total_cal = 0
    total_pro = 0

    food_list = list(foods.values())

    for i in range(len(chrom)):
        qty = chrom[i]
        total_cost += qty * food_list[i][0]
        total_cal += qty * food_list[i][1]
        total_pro += qty * food_list[i][2]

    penalty = 0

    if total_cal < TARGET_CALORIES:
        penalty += (TARGET_CALORIES - total_cal) * 10

    if total_pro < TARGET_PROTEIN:
        penalty += (TARGET_PROTEIN - total_pro) * 20

    return 1 / (1 + total_cost + penalty)

# Crossover
```

```
def crossover(p1, p2):
    child = []
    for i in range(len(p1)):
        child.append(random.choice([p1[i], p2[i]]))
    return child

# Mutation
def mutation(chrom):
    idx = random.randint(0, len(chrom) - 1)
    chrom[idx] = max(0, chrom[idx] + random.choice([-1, 1]))
    return chrom

# Initial population
population = [create_chromosome() for _ in range(POP_SIZE)]

for _ in range(GENERATIONS):
    population.sort(key=fitness, reverse=True)

    new_population = population[:2]  # elite

    while len(new_population) < POP_SIZE:
        p1, p2 = random.sample(population[:10], 2)
        child = crossover(p1, p2)

        if random.random() < 0.2:
            child = mutation(child)

        new_population.append(child)

    population = new_population

# Best solution
best = max(population, key=fitness)

print("Best Diet Plan")
food_names = list(foods.keys())

for i in range(len(best)):
    print(food_names[i], ":", best[i])

# Calculate totals
cost = calories = protein = 0

for i in range(len(best)):
    qty = best[i]
    cost += qty * foods[food_names[i]][0]
    calories += qty * foods[food_names[i]][1]
    protein += qty * foods[food_names[i]][2]
```

```
print("\nTotal Cost =", cost)
print("Calories =", calories)
print("Protein =", protein)
```

Output:

```
Best Diet Plan
Rice : 2
Milk : 0
Egg : 5
Banana : 3

Total Cost = 105
Calories = 1050
Protein = 41
```